

EXPERIMENT – 5

CODE:

```
int count = 0;

const byte segPins[] = {1,2,3,4,5,6,7};

// Common Cathode (a,b,c,d,e,f,g)
const byte digits[10][7] = {
  {1,1,1,1,1,1,0}, //0
  {0,1,1,0,0,0,0}, //1
  {1,1,0,1,1,0,1}, //2
  {1,1,1,1,0,0,1}, //3
  {0,1,1,0,0,1,1}, //4
  {1,0,1,1,0,1,1}, //5
  {1,0,1,1,1,1,1}, //6
  {1,1,1,0,0,0,0}, //7
  {1,1,1,1,1,1,1}, //8
  {1,1,1,1,0,1,1} //9
};

void displayDigit(int d)
{
  for(int i=0;i<7;i++)
    digitalWrite(segPins[i], digits[d][i]);
}

void setup()
{
```

```
pinMode(12, INPUT_PULLUP);
```

```
for(int i=1;i<=7;i++)
```

```
    pinMode(i, OUTPUT);
```

```
    displayDigit(0);
```

```
}
```

```
void loop()
```

```
{
```

```
    if(digitalRead(12)==LOW)
```

```
    {
```

```
        delay(200);
```

```
        count++;
```

```
        if(count>9)
```

```
            count=0;
```

```
        displayDigit(count);
```

```
        while(digitalRead(12)==LOW);
```

```
    }
```

```
}
```

OUTPUT:

